

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element C Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design requirements, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design requirements, be sure to delete the instructions. Make sure to list your design requirements in ranked order, with design requirement #1 having the highest priority. If possible, list 5 requirements that are labeled as HIGH priority and 1-5 additional requirements that are labeled as LOW priority.

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

Filter, have a filter to get the harmful smells and particulate out of the air.

This ties to our research because you have to have a filter to “grab” the harmful things

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #2 (HIGH Priority)

Have a fan to filter the air quickly.

This ties to our research because we have seen past examples with the fan for efficiency. Also needs a fan so that it can circulate the air and help the filter be more efficient and accurate. This could help with odors, smells, particles, and anything else in the air that

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needs to get filtered. Resin and other chemicals that are being released come in masses and don't stop until the material is either hardened or not under change.

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Design Requirement #3 (HIGH Priority)

It has to be mounted to "attack" the bad air faster because the scents rise.

This ties into our research because this helps take away the main thing Mrs. Poole asked because the harmful air particles rose. Mrs. Poole wants to ensure the safety of her students, that is her main priority. If it is also mounted it is a bonus, meaning it would take up less space; maximizing the area to create in.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #4 (HIGH Priority)

It needs to be light so it can be mounted.

This ties into our research because if it is mounted, it can take out the harmful things quicker. It is also a requirement designated by our customer (Mrs Poole) that the air purifier we build must be off the ground. Being light will lower the difficulty of achieving success.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #5 (HIGH Priority)

Buttons/ remote controller

Explain how this design requirement directly ties to your prior research.

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Design Requirement #6 (LOW Priority)

Batteries/cord

Explain how this design requirement directly ties to your prior research.

Mrs. Poole prefers a battery power air filter, in order to conserve energy. This will help keep the electricity bill low, and rechargeable batteries can be used.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #7 (LOW Priority)

Quiet

Explain how this design requirement directly ties to your prior research.

The ability to operate at a low volume will allow students to focus on their projects while the air is being purified of dangerous particulates.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #8 (VERY LOW Priority)

Has a Handle.

Explain how this design requirement directly ties to your prior research

It should have a handle to make it easier to carry around if it has to be moved.



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